

★ 28.2-5

Generalize the matrix-inversion algorithm of Theorem 28.2 to handle matrices of complex numbers, and prove that your generalization works correctly. (*Hint:* Instead of the transpose of A , use the *conjugate transpose* A^* , which you obtain from the transpose of A by replacing every entry with its complex conjugate. Instead of symmetric matrices, consider *Hermitian* matrices, which are matrices A such that $A = A^*$.)

28.3 Symmetric positive-definite matrices and least-squares approximation

Symmetric positive-definite matrices have many interesting and desirable properties. An $n \times n$ matrix A is *symmetric positive-definite* if $A = A^T$ (A is symmetric) and $x^T Ax > 0$ for all n -vectors $x \neq 0$ (A is positive-definite). Symmetric positive-definite matrices are nonsingular, and an LU decomposition on them will not divide by 0. This section proves these and several other important properties of symmetric positive-definite matrices. We'll also see an interesting application to curve fitting by a least-squares approximation.

The first property we prove is perhaps the most basic.

Lemma 28.3

Any positive-definite matrix is nonsingular.

Proof Suppose that a matrix A is singular. Then by Corollary D.3 on page 1221, there exists a nonzero vector x such that $Ax = 0$. Hence, $x^T Ax = 0$, and A cannot be positive-definite.

■

The proof that an LU decomposition on a symmetric positive-definite matrix A won't divide by 0 is more involved. We begin by proving properties about certain submatrices of A . Define the *k th leading submatrix* of A to be the matrix A_k consisting of the intersection of the first k rows and first k columns of A .

Lemma 28.4

If A is a symmetric positive-definite matrix, then every leading submatrix of A is symmetric and positive-definite.

Proof Since A is symmetric, each leading submatrix A_k is also symmetric. We'll prove that A_k is positive-definite by contradiction. If A_k is not positive-definite, then there exists a k -vector $x_k \neq 0$ such that $x_k^T A_k x_k \leq 0$. Let A be $n \times n$, and

$$A = \begin{pmatrix} A_k & B^T \\ B & C \end{pmatrix} \quad (28.16)$$

for submatrices B (which is $(n-k) \times k$) and C (which is $(n-k) \times (n-k)$). Define the n -vector $x = (x_k^T \ 0)^T$, where $n - k$ 0s follow x_k . Then we have

$$\begin{aligned} x^T A x &= (x_k^T \ 0) \begin{pmatrix} A_k & B^T \\ B & C \end{pmatrix} \begin{pmatrix} x_k \\ 0 \end{pmatrix} \\ &= (x_k^T \ 0) \begin{pmatrix} A_k x_k \\ B x_k \end{pmatrix} \\ &= x_k^T A_k x_k \\ &\leq 0, \end{aligned}$$

which contradicts A being positive-definite. ■

We now turn to some essential properties of the Schur complement. Let A be a symmetric positive-definite matrix, and let A_k be a leading $k \times k$ submatrix of A . Partition A once again according to equation (28.16). Equation (28.10) generalizes to define the **Schur complement** S of A with respect to A_k as

$$S = C - B A_k^{-1} B^T. \quad (28.17)$$

(By Lemma 28.4, A_k is symmetric and positive-definite, and therefore, A_k^{-1} exists by Lemma 28.3, and S is well defined.) The earlier definition (28.10) of the Schur complement is consistent with equation (28.17) by letting $k = 1$.

The next lemma shows that the Schur-complement matrices of symmetric positive-definite matrices are themselves symmetric and

positive-definite. We used this result in Theorem 28.2, and its corollary will help prove that LU decomposition works for symmetric positive-definite matrices.

Lemma 28.5 (Schur complement lemma)

If A is a symmetric positive-definite matrix and A_k is a leading $k \times k$ submatrix of A , then the Schur complement S of A with respect to A_k is symmetric and positive-definite.

Proof Because A is symmetric, so is the submatrix C . By Exercise D.2-6 on page 1223, the product $BA_k^{-1}B^T$ is symmetric. Since C and $BA_k^{-1}B^T$ are symmetric, then by Exercise D.1-1 on page 1219, so is S .

It remains to show that S is positive-definite. Consider the partition of A given in equation (28.16). For any nonzero vector x , we have $x^T Ax > 0$ by the assumption that A is positive-definite. Let the subvectors y and z consist of the first k and last $n - k$ elements in x , respectively, and thus they are compatible with A_k and C , respectively. Because A_k^{-1} exists, we have

$$\begin{aligned} x^T Ax &= (y^T \ z^T) \begin{pmatrix} A_k & B^T \\ B & C \end{pmatrix} \begin{pmatrix} y \\ z \end{pmatrix} \\ &= (y^T \ z^T) \begin{pmatrix} A_k y + B^T z \\ B y + C z \end{pmatrix} \\ &= y^T A_k y + y^T B^T z + z^T B y + z^T C z \\ &= (y + A_k^{-1} B^T z)^T A_k (y + A_k^{-1} B^T z) + z^T (C - B A_k^{-1} B^T) z, \end{aligned} \quad (28.18)$$

This last equation, which you can verify by multiplying through, amounts to “completing the square” of the quadratic form. (See Exercise 28.3-2.)

Since $x^T Ax > 0$ holds for any nonzero x , pick any nonzero z and then choose $y = -A_k^{-1} B^T z$, which causes the first term in equation (28.18) to vanish, leaving

$$z^T (C - B A_k^{-1} B^T) z = z^T S z$$

as the value of the expression. For any $z \neq 0$, we therefore have $z^T S z = x^T Ax > 0$, and thus S is positive-definite. ■

Corollary 28.6

LU decomposition of a symmetric positive-definite matrix never causes a division by 0.

Proof Let A be an $n \times n$ symmetric positive-definite matrix. In fact, we'll prove a stronger result than the statement of the corollary: every pivot is strictly positive. The first pivot is a_{11} . Let e_1 be the length- n unit vector $(1 \ 0 \ 0 \ \cdots \ 0)^T$, so that $a_{11} = e_1^T A e_1$, which is positive because e_1 is nonzero and A is positive definite. Since the first step of LU decomposition produces the Schur complement of A with respect to $A_1 = (a_{11})$, Lemma 28.5 implies by induction that all pivots are positive. ■

Least-squares approximation

One important application of symmetric positive-definite matrices arises in fitting curves to given sets of data points. You are given a set of m data points

$$(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m),$$

where you know that the y_i are subject to measurement errors. You wish to determine a function $F(x)$ such that the approximation errors

$$\eta_i = F(x_i) - y_i \tag{28.19}$$

are small for $i = 1, 2, \dots, m$. The form of the function F depends on the problem at hand. Let's assume that it has the form of a linearly weighted sum

$$F(x) = \sum_{j=1}^n c_j f_j(x),$$

where the number n of summands and the specific *basis functions* f_j are chosen based on knowledge of the problem at hand. A common choice is $f_j(x) = x^{j-1}$, which means that

$$F(x) = c_1 + c_2x + c_3x^2 + \dots + c_nx^{n-1}$$

is a polynomial of degree $n - 1$ in x . Thus, if you are given m data points $(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)$, you need to calculate n coefficients $c_1, c_2, \dots, c_n$ that minimize the approximation errors $\eta_1, \eta_2, \dots, \eta_m$.

By choosing $n = m$, you can calculate each y_i *exactly* in equation (28.19). Such a high-degree polynomial F “fits the noise” as well as the data, however, and generally gives poor results when used to predict y for previously unseen values of x . It is usually better to choose n significantly smaller than m and hope that by choosing the coefficients c_j well, you can obtain a function F that finds the significant patterns in the data points without paying undue attention to the noise. Some theoretical principles exist for choosing n , but they are beyond the scope of this text. In any case, once you choose a value of n that is less than m , you end up with an overdetermined set of equations whose solution you wish to approximate. Let’s see how to do so.

Let

$$A = \begin{pmatrix} f_1(x_1) & f_2(x_1) & \dots & f_n(x_1) \\ f_1(x_2) & f_2(x_2) & \dots & f_n(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ f_1(x_m) & f_2(x_m) & \dots & f_n(x_m) \end{pmatrix}$$

denote the matrix of values of the basis functions at the given points, that is, $a_{ij} = f_j(x_i)$. Let $c = (c_k)$ denote the desired n -vector of coefficients. Then,

$$\begin{aligned} Ac &= \begin{pmatrix} f_1(x_1) & f_2(x_1) & \dots & f_n(x_1) \\ f_1(x_2) & f_2(x_2) & \dots & f_n(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ f_1(x_m) & f_2(x_m) & \dots & f_n(x_m) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} \\ &= \begin{pmatrix} F(x_1) \\ F(x_2) \\ \vdots \\ F(x_m) \end{pmatrix} \end{aligned}$$

is the m -vector of “predicted values” for y . Thus,

$$\eta = Ac - y$$

is the m -vector of *approximation errors*.

To minimize approximation errors, let's minimize the norm of the error vector η , which gives a *least-squares solution*, since

$$\|\eta\| = \left(\sum_{i=1}^m \eta_i^2 \right)^{1/2}.$$

Because

$$\|\eta\|^2 = \|Ac - y\|^2 = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij}c_j - y_i \right)^2,$$

to minimize $\|\eta\|$, differentiate $\|\eta\|^2$ with respect to each c_k and then set the result to 0:

$$\frac{d\|\eta\|^2}{dc_k} = \sum_{i=1}^m 2 \left(\sum_{j=1}^n a_{ij}c_j - y_i \right) a_{ik} = 0. \quad (28.20)$$

The n equations (28.20) for $k = 1, 2, \dots, n$ are equivalent to the single matrix equation

$$(Ac - y)^T A = 0$$

or, equivalently (using Exercise D.1-2 on page 1219), to

$$A^T(Ac - y) = 0,$$

which implies

$$A^T Ac = A^T y. \quad (28.21)$$

In statistics, equation (28.21) is called the *normal equation*. The matrix $A^T A$ is symmetric by Exercise D.1-2, and if A has full column rank, then by Theorem D.6 on page 1222, $A^T A$ is positive-definite as well. Hence, $(A^T A)^{-1}$ exists, and the solution to equation (28.21) is

$$\begin{aligned} c &= ((A^T A)^{-1} A^T) y \\ &= A^+ y, \end{aligned} \quad (28.22)$$

where the matrix $A^+ = ((A^T A)^{-1} A^T)$ is the *pseudoinverse* of the matrix A . The pseudoinverse naturally generalizes the notion of a matrix

inverse to the case in which A is not square. (Compare equation (28.22) as the approximate solution to $Ac = y$ with the solution $A^{-1}b$ as the exact solution to $Ax = b$.)

As an example of producing a least-squares fit, suppose that you have five data points

$$(x_1, y_1) = (-1, 2),$$

$$(x_2, y_2) = (1, 1),$$

$$(x_3, y_3) = (2, 1),$$

$$(x_4, y_4) = (3, 0),$$

$$(x_5, y_5) = (5, 3),$$

shown as orange dots in Figure 28.3, and you want to fit these points with a quadratic polynomial

$$F(x) = c_1 + c_2x + c_3x^2.$$

Start with the matrix of basis-function values

$$A = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \\ 1 & x_5 & x_5^2 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 5 & 25 \end{pmatrix},$$

whose pseudoinverse is

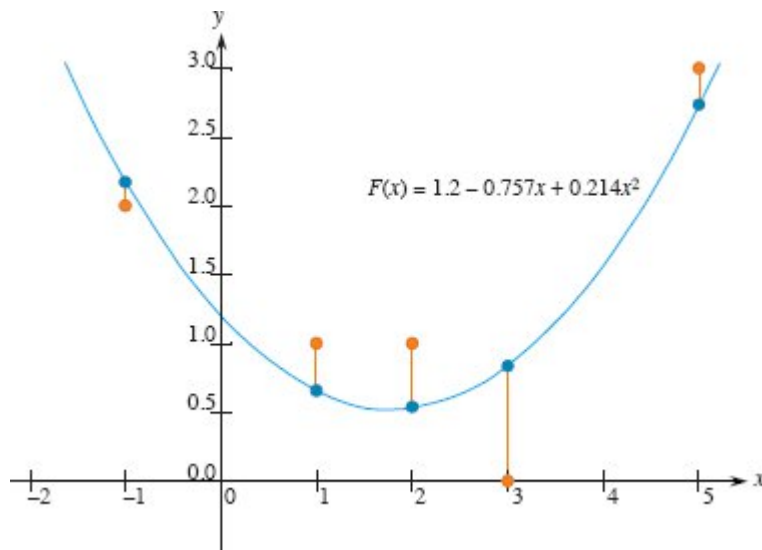


Figure 28.3 The least-squares fit of a quadratic polynomial to the set of five data points $\{(-1, 2), (1, 1), (2, 1), (3, 0), (5, 3)\}$. The orange dots are the data points, and the blue dots are their estimated values predicted by the polynomial $F(x) = 1.2 - 0.757x + 0.214x^2$, the quadratic polynomial that minimizes the sum of the squared errors, plotted in blue. Each orange line shows the error for one data point.

$$A^+ = \begin{pmatrix} 0.500 & 0.300 & 0.200 & 0.100 & -0.100 \\ -0.388 & 0.093 & 0.190 & 0.193 & -0.088 \\ 0.060 & -0.036 & -0.048 & -0.036 & 0.060 \end{pmatrix}.$$

Multiplying y by A^+ gives the coefficient vector

$$c = \begin{pmatrix} 1.200 \\ -0.757 \\ 0.214 \end{pmatrix},$$

which corresponds to the quadratic polynomial

$$F(x) = 1.200 - 0.757x + 0.214x^2$$

as the closest-fitting quadratic to the given data, in a least-squares sense.

As a practical matter, you would typically solve the normal equation (28.21) by multiplying y by A^T and then finding an LU decomposition of $A^T A$. If A has full rank, the matrix $A^T A$ is guaranteed to be

nonsingular, because it is symmetric and positive-definite. (See Exercise D.1-2 and Theorem D.6.)

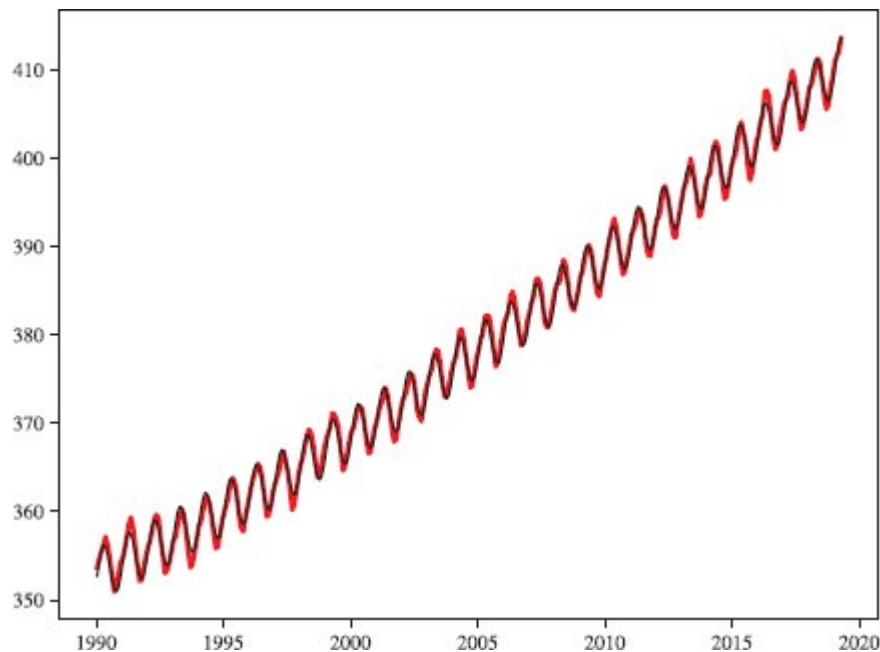


Figure 28.4 A least-squares fit of a curve of the form

$$c_1 + c_2x + c_3x^2 + c_4 \sin(2\pi x) + c_5 \cos(2\pi x)$$

for the carbon-dioxide concentrations measured in Mauna Loa, Hawaii from 1990¹ to 2019, where x is the number of years elapsed since 1990. This curve is the famous “Keeling curve,” illustrating curve-fitting to nonpolynomial formulas. The sine and cosine terms allow modeling of seasonal variations in CO_2 concentrations. The red curve shows the measured CO_2 concentrations. The best fit, shown in black, has the form

$$352.83 + 1.39x + 0.02x^2 + 2.83 \sin(2\pi x) - 0.94 \cos(2\pi x).$$

We close this section with an example in Figure 28.4, illustrating that a curve can also fit a nonpolynomial function. The curve confirms one aspect of climate change: that carbon dioxide (CO_2) concentrations have steadily increased over a period of 29 years. Linear and quadratic terms model the annual increase, and sine and cosine terms model seasonal variations.

Exercises

28.3-1

Prove that every diagonal element of a symmetric positive-definite matrix is positive.

28.3-2

Let $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ be a 2×2 symmetric positive-definite matrix. Prove that its determinant $ac - b^2$ is positive by “completing the square” in a manner similar to that used in the proof of Lemma 28.5.

28.3-3

Prove that the maximum element in a symmetric positive-definite matrix lies on the diagonal.

28.3-4

Prove that the determinant of each leading submatrix of a symmetric positive-definite matrix is positive.

28.3-5

Let A_k denote the k th leading submatrix of a symmetric positive-definite matrix A . Prove that $\det(A_k)/\det(A_{k-1})$ is the k th pivot during LU decomposition, where, by convention, $\det(A_0) = 1$.

28.3-6

Find the function of the form

$$F(x) = c_1 + c_2 x \lg x + c_3 e^x$$

that is the best least-squares fit to the data points

(1, 1), (2, 1), (3, 3), (4, 8).

28.3-7

Show that the pseudoinverse A^+ satisfies the following four equations:

$$AA^+A = A,$$

$$A^+AA^+ = A^+,$$

$$(AA^+)^\mathsf{T} = AA^+,$$

$$(A^+A)^\mathsf{T} = A^+A.$$

Problems

28-1 *Tridiagonal systems of linear equations*

Consider the tridiagonal matrix

$$A = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}.$$

- a.** Find an LU decomposition of A .
- b.** Solve the equation $Ax = (1\ 1\ 1\ 1\ 1)^\mathsf{T}$ by using forward and back substitution.
- c.** Find the inverse of A .
- d.** Show how to solve the equation $Ax = b$ for any $n \times n$ symmetric positive-definite, tridiagonal matrix A and any n -vector b in $O(n)$ time by performing an LU decomposition. Argue that any method based on forming A^{-1} is asymptotically more expensive in the worst case.
- e.** Show how to solve the equation $Ax = b$ for any $n \times n$ nonsingular, tridiagonal matrix A and any n -vector b in $O(n)$ time by performing an LUP decomposition.

28-2 *Splines*

A practical method for interpolating a set of points with a curve is to use **cubic splines**. You are given a set $\{(x_i, y_i) : i = 0, 1, \dots, n\}$ of $n + 1$ point-value pairs, where $x_0 < x_1 < \dots < x_n$. Your goal is to fit a piecewise-cubic curve (spline) $f(x)$ to the points. That is, the curve $f(x)$ is made up of n cubic polynomials $f_i(x) = a_i + b_ix + c_ix^2 + d_ix^3$ for $i =$